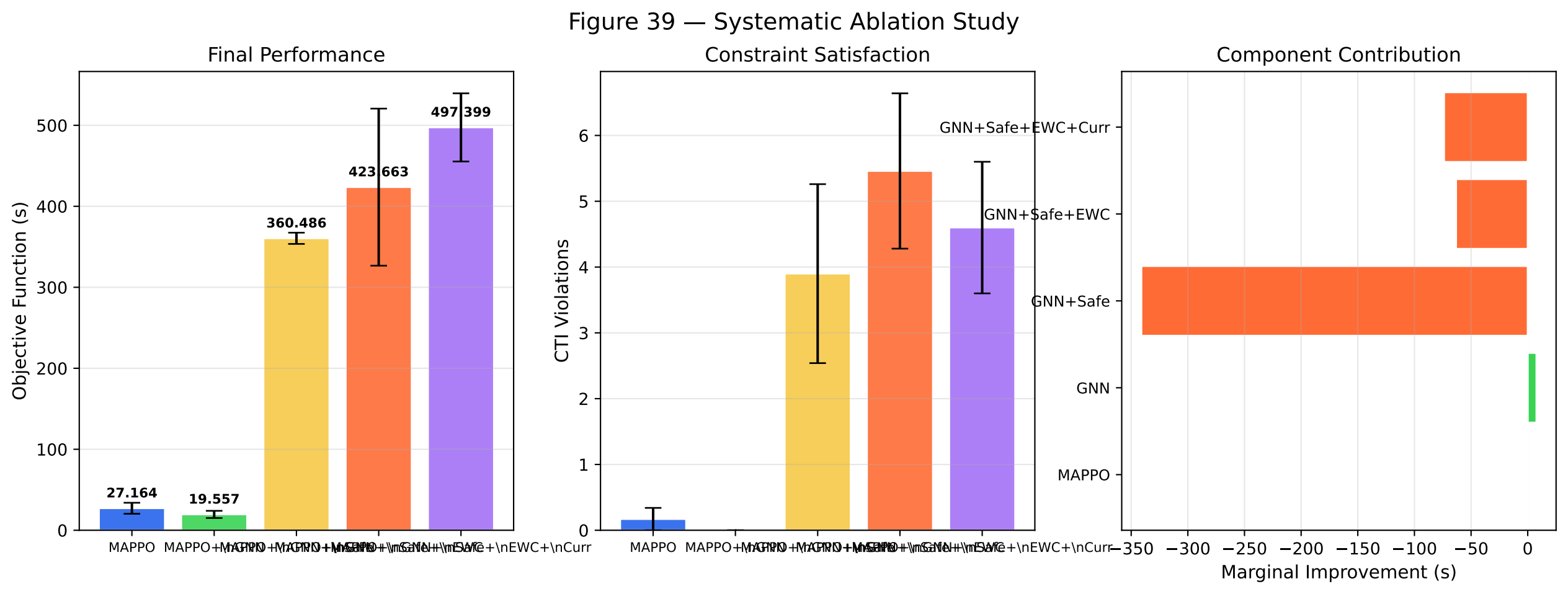
Figure 39 — Systematic Ablation Study
Final Performance
Objective Function (s)
0
100
200
300
400
500
27.164
19.557
360.486
423 663
497 399
MAPPO
MAPPO+\nGNN
MAPPO+\nSafe
MAPPO+\nSafe+\nEWC
MAPPO+\nSafe+\nEWC+\nCurr
Constraint Satisfaction
CTI Violations
0
1
2
3
4
5
6
MAPPO
MAPPO+\nGNN
MAPPO+\nSafe
MAPPO+\nSafe+\nEWC
MAPPO+\nSafe+\nEWC+\nCurr
Component Contribution
GNN+Safe+EWC+Curr
GNN+Safe+EWC
GNN+Safe
GNN
MAPPO
−350
−300
−250
−200
−150
−100
−50
0
Marginal Improvement (s)